

ALONSO MARCOS MUÑOZ

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Professional profile

Data Engineer focused on data governance, data quality and interoperability in the public sector. I deploy multi-version CKAN (PostgreSQL/Redis/Solr) with multi-tenant nginx routing, evolve DCAT/schemingdcat/harvest/spatial extensions and custom plugins, and industrialize DCAT-AP-ES compliant metadata ETL pipelines (Python, RDF/JSON-LD, SHACL, advanced SQL on PostgreSQL/Oracle) backing 10,000+ federated datasets in datos.gob.es for 22 Spanish State Administration (AGE) organizations; AI harness engineering with agentic AI (Claude, Codex, Copilot Pro) – subagents, worktrees and context management – across development, pipeline supervision, e2e tests and code audit.

Professional experience

Tragsatec – Data Analyst Programmer

Albacete, Spain | Jun. 2025 – Present

ImpulsaDATA / Public Data Demand Management Project (MRR – Spanish State Administration) / 15-person platforms team within a ~80-person project

- End-to-end metadata conversion ETL pipeline (Python, JSON-LD, DCAT-AP-ES, RDF, SHACL/pySHACL) with semantic validation; backs 10,000+ federated datasets in datos.gob.es for 22 AGE organizations.
- Multi-version CKAN (internal PostgreSQL/Redis/Solr); ckanext-dcat (common DCAT-AP-ES RDF profile), schemingdcat (custom schema), harvest, spatial (GeoJSON), and a custom plugin; Jinja2 over Action API.
- SQL modeling on the metadata model (PostgreSQL/Oracle): transformations, aggregations, views and analytical query optimization over 10,000+ datasets.
- Docker, Docker Compose, and Podman; nginx (reverse proxy) with virtual hosts/DNS per organization for the 22 multi-tenant catalogs; SSH administration across test/dev/pre/prod; zero H/VH/C vulnerabilities in shared deliverables.
- AI harness engineering with agentic AI (Claude/Claude Code, Codex, GitHub Copilot Pro): subagents, worktrees and context/token management applied to development, pipeline supervision, e2e tests and code audit; informal technical ownership and mentoring of 4 intern/junior team members.

La Fábrica del Tiempo – Junior Process Automation & Productivity Consultant (Internship) Almansa, Spain | Feb. 2024 – Aug. 2024

Power Platform (Power Apps + Power Automate) on Microsoft 365/SharePoint: internal app and no-code flows for CRM lead management, SharePoint–CRM sync and automated notifications/approvals; >25% operational efficiency gains over the ADOCC methodology.

City Council of Alcázar de San Juan – IT Technician (Internship)

Mar. 2019 – Jun. 2019

Selected projects

- **Master's thesis** (UCLM, in progress): metadata model in OpenMetadata aligned with DCAT-AP-ES (DCAT-AP 2.1.1 + HVD 2.2.0) with K8s/Helm, Python ingestor, JSON-LD, SHACL, and a Next.js console – github.com/AlonsoMarcosM/TFM_AlonsoM99.
- **Smart Parking Albacete** (UCLM, MUBDCN): serverless IoT platform in AWS with 40 MQTT/TLS sensors over IoT Core, Lambda + on-demand DynamoDB, REST API with GeoJSON, and Streamlit dashboard – github.com/AlonsoMarcosM/smart-parking-albacete.
- **Telco Churn MLOps** (UCLM, MUBDCN): Databricks Lakehouse with Auto Loader, Medallion on Delta Lake, MLflow, champion/challenger governance in Unity Catalog, and drift monitoring – github.com/AlonsoMarcosM/Trabajo-...-BIG-DATA.

Education

UCLM – Master's Degree in Big Data and Cloud Computing (in progress, expected Jun. 2026)

Sep. 2025 – Jun. 2026

UCLM – Bachelor's Degree in Computer Engineering, specialization in Information Technology

Sep. 2019 – Jun. 2025

IES Juan Bosco – Advanced Vocational Diploma in Network Systems Administration

Sep. 2017 – Jun. 2019

Certifications

CAPM – Project Management Institute – Certified foundations in PMI project management.

First Certificate in English (B2) – Cambridge English – Certified professional working proficiency in English.

Skills

Data Governance & Quality: data governance, data quality, metadata management, lineage, DCAT-AP-ES, RDF, JSON-LD, SHACL, pySHACL, OpenMetadata.

Data Engineering: Python, ETL, CKAN + extensions (dcat/schemingdcat/harvest/spatial), PostgreSQL, Oracle, Redis, Solr, advanced SQL, REST APIs, Jinja2.

Cloud & DevOps: Docker, Docker Compose, Podman, nginx, Linux, SSH, Kubernetes, Helm, AWS, Terraform, GitLab, GitHub, Git.

Agentic AI / AI harness engineering: Claude/Claude Code, Codex, GitHub Copilot Pro; subagents, worktrees, context/token management, AI-assisted e2e testing and code audit.

Big Data: Airflow, Spark, Kafka, Delta Lake, MinIO, SQL Server, Databricks, MLflow.

Power Platform & other: Power Apps, Power Automate, Microsoft 365/SharePoint, Angular, TypeScript.

Management: PMI/CAPM, Kanban, ADOCC.

Languages: native Spanish, English B2 (Cambridge First).